Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/29/2006
PI Name:	Murnane, John Ph.D.		
Project Title:	Telomere Repair and Chromosome Instability After High LET Particle Radiation		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94103	**Congressional District:**	43
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	01-OBPR-07
Start Date:	10/01/2003	**End Date:**	07/31/2005
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG9-1261		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Chromosome instability plays an important role in human cancer by providing the genome plasticity required for tumor cell progression. In this proposal, we will show that spontaneous telomere loss is a mechanism for initiating chromosome instability in human cells. We will further present data that show the effectiveness of high LET particles, a component of the space radiation environment, in generating non-rejoined DNA double-strand breaks (DSBs) in the genome of human cells. Based on these preliminary results, we present the hypothesis that high LET particles generate DNA DSBs in the telomeric regions of the genome and that a fraction of these breaks do not rejoin. The loss of the ends of chromosomes due to these un-rejoined DSBs may not be lethal to the cell, but may instead result in the loss of functional telomeres, chromosome fusion, and initiation of breakage/fusion/bridge cycle-induced chromosome instability. Therefore, this study will present a molecular mechanism for the initiation of high LET particle-induced chromosome instability in human cells and hence a better understanding of the factors involved in predicting carcinogenesis in humans during space flight. To test our hypothesis, this proposal contains three specific aims: 1) to determine the rate of		

	telomere loss, spectrum of events following telomere loss and consequences of telomere loss after high-LET particle irradiation in a human squamous cell carcinoma cell line containing a selectable marker integrated directly adjacent to a telomere, 2) to measure the rate of DSB induction and capacity to repair those DSBs in telomeric regions of the genome after high and low LET radiation using a modified pulsed-field gel electrophoresis (PFGE) assay and 3) to extend these studies to cell lines that are normal in their cell cycle control checkpoints, including telomerase immortalized primary human fibroblasts and mouse embryonic stem (ES) cells with which an animal model containing a marked telomere can be developed.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	New project for FY2004.
Bibliography Type:	Description: (Last Updated:)